

When are Imprecise Bandits Computationally Tractable?

author names withheld

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1. Setting

An instance of the imprecise bandits problem is a tuple $\mathcal{M} = (X, D, H, r)$, where X is a set of arms, D is a set of possible outcomes, H is a set of mappings of type $X \rightarrow \square D$ where $\square D$ denotes the set of credal sets over D , and $r : X \times D \rightarrow \mathbb{R}$ is a reward function.

The imprecise bandits game is played between a learner and an adversary for T rounds (both adversary and learner know T). At the beginning of the game, the adversary picks some $h^* \in H$. The learner knows X, D, H, r , and T , but does not have knowledge of h^* . Then, at each round t :

1. Learner picks an arm $x_t \in X$.
2. Adversary picks a distribution $P_t \in h^*(x_t)$.
3. An outcome $y_t \sim P_t$ is drawn and shown to the learner as feedback.
4. Learner gets reward $r(x_t, y_t)$.

Given a hypothesis $h \in H$, we define the value functions:

$$v_h(x) := \min_{P \in h(x)} \mathbb{E}_{y \sim P} [r(x, y)], \quad V_h := \max_{x \in X} v_h(x).$$

We use the shorthand $v^*(x)$ and V^* to denote the value functions corresponding to the true hypothesis h^* .

The regret over horizon T is

$$R_T := TV^* - \mathbb{E} \left[\sum_{t=1}^T r(x_t, y_t) \right].$$

Here the expectation in the second term is taken wrt the randomness in the learner's and the adversary's strategies as well as the randomness arising from sampling $y_t \sim P_t$.

We are interested in learner strategies that can achieve a sublinear (in T) regret for all true hypotheses $h^* \in H$. We say that an imprecise bandits instance $\mathcal{M} = (X, D, H, r)$ has a *statistically efficient* learner if such a learner strategy exists.

Computational tractability. In this paper, we are interested in studying the computational complexity of the learner. A *learner policy* π is a randomized algorithm that takes as input a history $h_t = ((x_i, y_i))_{i=1}^t$ and the description of the problem instance $\mathcal{M}$ and returns a next arm x_{t+1} . The computational complexity of the problem depends crucially on the exact nature of access the policy is given to $\mathcal{M}$. For example, on one extreme, one can consider a non-uniform access such that a policy $\pi^{\mathcal{M}}$ is merely indexed by the instance $\mathcal{M}$ as opposed to $\mathcal{M}$ being an input to a computational

procedure. However, this makes several instances rather trivial. In this paper we mostly consider specific encodings of $\mathcal{M}$. We also occasionally assume π to have oracle access to $\mathcal{M}$ with certain specific oracles. We will make these assumptions clear in the respective sections.

We will say that a learner is *computationally efficient* if it runs in time polynomial in the size of its input and achieves a regret $R_T \leq \text{poly}(|\mathcal{M}|) T^{1-\alpha}$ for some $\alpha > 0$.

1.1. Linear Imprecise Bandits

For most of this work, we study the following linear specialization of the instance $\mathcal{M} = (X, D, H, r)$ defined above. Let $X \subseteq \mathbb{R}^{d_X}$ be the arm set, and let $D \subseteq \mathbb{R}^{d_D}$ be the outcome set. We assume that X and D are Euclidean balls of radii R_X and R_D (not necessarily centred at the origin). The reward is affine in the arm and outcome:

$$r(x, y) := a^T x + b^T y + c.$$

Let the hypothesis class H be parametrized by $Z \subseteq \mathbb{R}^{d_Z}$. For every $z \in Z$, we define the hypothesis h_z by first defining the possible expected values of the distributions that h_z is allowed to choose. In particular, let

$$K_z(x) := \{y \in D : C_z y + B_z x + d_z = 0\}. \quad (1)$$

Here B_z, C_z , and d_z are matrices and vectors of appropriate dimensions that depend affinely on z . Now, hypothesis $h_z : X \rightarrow \square D$ is given by

$$h_z(x) := \{P \in \Delta D : \mathbb{E}_{y \sim P}[y] \in K_z(x)\}.$$

Let $z^* \in Z$ represent the true hypothesis, so that $h^* = h_{z^*}$. The true feasible mean set is $K^*(x) := K_{z^*}(x)$.

We also assume a **uniform non-tangency condition**: there is a constant $S \in (0, 1]$ such that, for every $x \in X$ and every $p \in \mathbb{R}^{d_D}$ satisfying $C_{z^*} p + B_{z^*} x + d_{z^*} = 0$ with $p \notin D$,

$$\text{dist}(p, D) \geq S \text{dist}(p, K^*(x)).$$

For a Euclidean ball, this says that the affine solution spaces defined by the true constraints remain uniformly bounded away from tangency to D . This is the transversality condition used in [1].

Because r is affine in its outcome argument, the value functions defined above satisfy

$$v^*(x) = \min_{P \in h^*(x)} \mathbb{E}_{y \sim P}[r(x, y)] = \min_{y \in K^*(x)} r(x, y), \quad V^* := \max_{x \in X} v^*(x).$$

The regret over horizon T is

$$R_T := TV^* - \mathbb{E} \left[\sum_{t=1}^T r(x_t, y_t) \right].$$

To define the computational problem, we fix a finite representation scheme for the data defining $\mathcal{M}$, and let $|\mathcal{M}|$ denote the length of its description. We represent X and D by their centres and radii,

r by its coefficients, and H by a description of Z together with the affine map $z \mapsto (B_z, C_z, d_z)$. We restrict attention to succinctly represented instances, for which $|\mathcal{M}| \leq \text{poly}(d_X, d_D, d_Z)$.

Now we state our main result.

Theorem 1 (efficient $\sqrt{T}$ learning)

There exists a policy for linear imprecise bandits as defined above that, for every time horizon T and every true hypothesis $h^ \in H$, runs in time polynomial in $|\mathcal{M}|$ and T and achieves regret*

$$R_T \leq \tilde{O}(\sqrt{T}).$$

The hidden factor depends polynomially on the dimensions, S^{-1} , and the reward range. The precise dependence is given in Theorem 10.

This dependence on T is tight up to logarithmic factors: we can reuse the $\Omega(\sqrt{T})$ worst-case lower bound for stochastic bandits [2, Theorem 5.1]. To see this, take $X = [-1, 1]$, $D = [0, 1]$, $r(x, y) = y$, and, for unknown $\mu = (\mu_1, \mu_2) \in [0, 1]^2$, set

$$K_\mu(x) = \frac{\{1-x\}}{2}\mu_1 + \frac{1+x}{2}\mu_2.$$

These are affine constraints of the required form, with $S = 1$. Given a two-armed Bernoulli bandit with means μ_1 and μ_2 , we can simulate a play of x by pulling its first arm with probability $\frac{1-x}{2}$ and its second arm otherwise, and returning the observed reward as y . This gives a compatible outcome distribution at x and preserves expected regret, since $V^* = \max(\mu_1, \mu_2)$. Thus a better worst-case regret bound here would give a better bound for two-armed stochastic bandits.

2. Warmup: A $T^{\frac{2}{3}}$ Learner

For an arm x and outcome y , consider the vector $w(x, y) = (1, x, y)$. This vector lies in $\mathbb{R}^{1+d_X+d_D}$, and the constraints defining the feasible set of outcomes, $C_{z^*}y + B_{z^*}x + d_{z^*} = 0$, determine a linear subspace of this vector space. Define $\mathcal{L} := \ker(d_{z^*} \ B_{z^*} \ C_{z^*})$. A conditional mean m is feasible at x precisely when $m \in D$ and $w(x, m) \in \mathcal{L}$. The learner's task is to learn enough about $\mathcal{L}$ to get low regret. Of course, we may never learn all of $\mathcal{L}$. For example, if nature keeps choosing conditional means in a strictly smaller subspace, then we cannot distinguish that subspace from the true $\mathcal{L}$. But this is fine, since we do not need to learn directions that nature never uses.

The fact that we never get to observe the true conditional mean, but only a noisy observation of it, adds an extra complication. In a noiseless world, a new observation tells us a new direction of $\mathcal{L}$. Thus we can simply maintain a linear subspace of $\mathbb{R}^{1+d_Z+d_D}$, and incrementally grow it every time we get new information. However, in the presence of noise, no observation can be used to add an entire direction with certainty. Instead, we encode our knowledge of $\mathcal{L}$ with an *ellipsoid*, and incrementally grow it as we get new information. The details are written below.

We first translate and rescale the balls so that $X = \{x : \|x\|_2 \leq 1\}$ and $D = \{y : \|y\|_2 \leq 1\}$. We continue to write the reward as $r(x, y) = a^T x + b^T y + c$ and the true subspace as $\mathcal{L}$ in these coordinates. The new $\|b\|_2$ equals $R_D \|b\|_2$ in the original coordinates, and the non-tangency constant S stays the same. Write $q := 1 + d_X + d_D$ for the number of coordinates of $w(x, y)$. For a positive-definite matrix V , write $\|w\|_{V^{-1}} := \sqrt{w^T V^{-1} w}$.

Define $p_V(x, y) := \|w(x, y)\|_{V^{-1}}^2$, and $E_V := \left\{ w \in \mathbb{R}^q : \|w\|_{V^{-1}}^2 \leq 1 \right\}$.

Here E_V is the ellipsoid we described above, which we incrementally expand as we get new information about $\mathcal{L}$. The algorithm works in blocks of length n (we will choose n later to get a small regret). At the beginning of each block, we choose an arm x and play the same arm throughout the block. The ellipsoid E_V stays fixed during these rounds.

To choose an arm, we use our current guess for the feasible conditional means. For each x , this consists of the $m \in D$ with $(1, x, m) \in E_V$. If some arm has no such m , we choose it to learn more about it. Otherwise, we choose the arm with the largest worst-case reward over its guessed feasible means.

At the end of a full block, we average its n observations to get $\bar{y}$. We then check whether $(1, x, \bar{y})$ lies in E_V . If it does not, we enlarge the ellipsoid to include it. Otherwise, we leave the ellipsoid unchanged.

For the time being, we assume that we can carry out the rule for choosing an arm exactly. We will consider the computational aspects of the algorithm later.

Algorithm 1.

Fix a block length $n \in \{1, \dots, T\}$ and initialize $V := n^{-1}I_q$. At the start of each block:

1. For each arm, let

$$\hat{K}_V(x) := \{y \in D : p_V(x, y) \leq 1\}.$$

If any of these sets is empty, choose an arm x with $\hat{K}_V(x) = \emptyset$. Otherwise, define

$$H_V(x) := \min_{y \in \hat{K}_V(x)} r(x, y),$$

and choose

$$x \in \operatorname{argmax}_{x' \in X} H_V(x').$$

2. Play x for n rounds, or until the horizon if fewer rounds remain. Let $\bar{y}$ be the empirical mean of observed outcomes and $\bar{w} := w(x, \bar{y})$.
 3. If the block has length n and $p_V(x, \bar{y}) > 1$, update $V \leftarrow V + \bar{w}\bar{w}^T$. Otherwise leave V unchanged.
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Theorem 2 (Informal)

For fixed problem parameters, Algorithm 1 with exact planning has expected regret

$$R_T \leq \tilde{O}\left(n + \frac{T}{\sqrt{n}}\right).$$

In particular, choosing $n = \left\lceil T^{\frac{2}{3}} \right\rceil$ gives $R_T \leq \tilde{O}\left(T^{\frac{2}{3}}\right)$.

Proof sketch We will show that, with high probability, throughout the T time steps, every point in E_V is at most about $\frac{1}{\sqrt{n}}$ away from the true $\mathcal{L}$ (Lemma 4). We ignore constants depending on the fixed problem parameters and logarithmic factors in this sketch.

Now consider an uninformative block, meaning a full block in which we do not update V . The arm x that we play during this block is picked by maximizing H_V . Since the block is uninformative, its empirical mean outcome satisfies $w(x, \bar{y}) \in E_V$. These two facts, together with the distance bound and Lemma 2, imply that the average reward is at least V^* minus about $\frac{1}{\sqrt{n}}$. Thus the block's regret (the best arm's guaranteed reward nV^* minus the reward we actually achieved) is at most about $\frac{n}{\sqrt{n}} = \sqrt{n}$.

On the other hand, for an informative block, the empirical mean can lie outside the ellipsoid, so we only have the trivial regret bound of $O(n)$.

Finally, the update rule for V ensures that there are at most $O(\log T)$ informative blocks: each update more than doubles the determinant, whose total growth is bounded by a polynomial in T (see the proof of Theorem 9). There are at most $\frac{T}{n}$ uninformative blocks, so the total regret is about $n \log T + \sqrt{n} \frac{T}{n}$.

Ignoring logarithmic factors, this bound is minimized by choosing $n = \left\lceil T^{\frac{2}{3}} \right\rceil$, giving an upper bound of $\tilde{O}\left(T^{\frac{2}{3}}\right)$ on the regret. $\blacksquare$

The precise statement and proof are in Appendix C.

2.1. Computational tractability

The matrix updates and the test $p_V(x, \bar{y}) > 1$ can be computed in polynomial time. The main issue is choosing an arm, since it involves solving a min-max optimization.

Exact emptiness tests can be unstable when the two constraints just touch. To allow some numerical slack, the implementation uses

$$\hat{K}_V^{\text{in}}(x) := \{y \in D : p_V(x, y) \leq 4\}, \quad \hat{K}_V^{\text{out}}(x) := \{y \in D : p_V(x, y) \leq 9\}.$$

It looks for an arm whose inner set is empty, and otherwise plans against the outer sets. The update rule still uses $p_V > 1$. Thus we only change the arm-selection rule by constant factors in the ellipsoid's radius.

To find an empty inner set, let $d_V(x)$ be the Euclidean distance between $2E_V$ and $\{(1, x, y) : y \in D\}$. Writing $z = (z_0, z_X, z_D)$, the support functions of the ellipsoid and the unit ball give

$$d_V(x) = \max_{\|z\|_2 \leq 1} \left[z_0 + z_X^T x - 2\sqrt{z^T V z} - \|z_D\|_2 \right].$$

In particular, $d_V(x) > 0$ certifies that the inner set is empty. Maximize this expression jointly over $x \in X$ and z , to additive accuracy $\eta := \frac{1}{10n}$. If the returned lower bound exceeds η , play the corresponding arm. Otherwise $d_V(x) \leq 2\eta$ for every arm. The gap between $2E_V$ and $3E_V$ then guarantees that every outer set is nonempty, with enough room to carry out the remaining optimization.

Define the value used for planning by

$$H_V^{\text{out}}(x) := \min_{y \in \hat{K}_V^{\text{out}}(x)} r(x, y).$$

Convex duality gives

$$H_V^{\text{out}}(x) = a^T x + c + \max_{\|z\|_2 \leq M} \left[z_0 + z_X^T x - 3\sqrt{z^T V z} - \|b + z_D\|_2 \right],$$

where $M := 4n(1 + \|b\|_1)$ suffices. The strict margin obtained above is what lets us bound the dual variable uniformly over all arms.

Both maximizations reduce to a quadratic program with a fixed number of quadratic constraints. For example, replace the two norms in the last display by nonnegative variables t, s satisfying $z^T V z \leq t^2$ and $\|b + z_D\|_2^2 \leq s^2$. The objective becomes quadratic in (x, z, t, s) . All variables have explicit bounds, so we can also add a redundant enclosing ball. Bienstock’s theorem [3], stated in Appendix A, solves these problems to weak additive accuracy in polynomial time. The dual formulas, the bound on M , and the conversion of a weak solution into a valid arm and a certified lower bound are justified in Appendix C.A.

The regret argument still works. An empty inner set forces an update, whereas an uninformative block has its empirical mean in the outer set. Points in $3E_V$ are at most three times as far from $\mathcal{L}$ as the earlier bound for E_V . Thus the two types of blocks still cost $\tilde{O}(n)$ and $\tilde{O}(\sqrt{n})$, respectively. Maximizing H_V^{out} to additive accuracy ε adds at most $T\varepsilon$ to the regret. Taking $\varepsilon = (T+1)^{-2}$, and rounding the observations to the precision specified in the appendix, gives a polynomial-time implementation with the same $\tilde{O}(T^{\frac{2}{3}})$ rate.

3. A $\sqrt{T}$ learner

The previous section attempts to apply the principle of “optimism in the face of uncertainty,” but does not fully embrace it. Indeed, the arm for each block is picked optimistically, but each block still looks like a standard exploration block. The learner does not always need to wait for n steps before identifying that the arm it is playing is problematic. If the outcomes collected so far have an empirical mean $\bar{y}$ with $w(x, \bar{y})$ already far outside the current ellipsoid, then the block can be cut short more quickly. However, the current arm score does not account for outcomes outside the ellipsoid. First, we design a score function that takes every outcome in D into account, not just those represented inside our chosen ellipsoid. The score function will be linked to the regret in a way that lets us decide when to move on to the next block.

We construct this score by replacing the hard constraint $p_V(x, y) \leq 1$ in H_V with a penalty, so that we can minimize over all of D . For a parameter $\lambda > 0$, which we will choose later to get a small regret, define

$$U_V(x) := \min_{y \in D} [r(x, y) + \lambda p_V(x, y)].$$

We pick the arm x by maximizing U_V . For now, assume that we can maximize the score exactly.

The following lemma gives us a way to control regret when the outcome y appears anywhere in D .

Lemma 1 *Fix $\delta \in (0, 1)$. With probability at least $1 - \delta$, simultaneously for every matrix V at the start of a block of Algorithm 2, every $\lambda > 0$, and every arm x maximizing U_V with this value of λ ,*

$$V^* - r(x, y) \leq \lambda p_V(x, y) + \tilde{O}(1/\lambda) \quad \text{for every } y \in D.$$

Here $\tilde{O}$ hides logarithmic factors in T/δ .

Proof On the event of Lemma 3, the distance bound in Equation 6, proved in Appendix D, and Lemma 2 give, for every arm x' and outcome $y \in D$,

$$r(x', y) \geq v^*(x') - \tilde{O}\left(\sqrt{p_V(x', y)}\right).$$

The penalty grows quadratically in $\sqrt{p_V(x', y)}$, while the error bound grows linearly. In particular, $\lambda p - \tilde{O}(\sqrt{p}) \geq -\tilde{O}(1/\lambda)$ for every $p \geq 0$ and every $\lambda > 0$. Adding the penalty and minimizing over y therefore gives $U_V(x') \geq v^*(x') - \tilde{O}(1/\lambda)$. Since x maximizes the score, $U_V(x) \geq V^* - \tilde{O}(1/\lambda)$. Finally, the definition of the score gives $U_V(x) \leq r(x, y) + \lambda p_V(x, y)$ for every $y \in D$. Combining these bounds proves the claim. Since the distance bound is independent of λ , the argument holds for all $\lambda > 0$ on the same event. $\blacksquare$

To use this bound during a block, we keep V fixed and play the same arm x repeatedly, but recompute the empirical mean after every observation. Let $\bar{y}_n$ be the mean of the first n outcomes in the block. Since the reward is affine, the reward accumulated so far is $nr(x, \bar{y}_n)$. Applying Lemma 1 to this mean gives

$$nV^* - \sum_{i=1}^n r(x, y_i) \leq \lambda n p_V(x, \bar{y}_n) + \tilde{O}(n/\lambda).$$

The first term is observable. We can compute it after every observation, and we end the block and update V the first time it reaches λ , or equivalently, when $n p_V(x, \bar{y}_n) \geq 1$. Before this happens, the block's regret is bounded by $\lambda + \tilde{O}(n/\lambda)$.

A large empirical penalty can therefore make us update after only a few observations. A small one lets us keep playing the arm for longer. The factor n accounts for how long we have been playing: we may need to update even while the empirical mean satisfies the ellipsoid constraint, because its penalty has been multiplied by a large block length. Thus we choose when to update V by monitoring this bound throughout the block, rather than waiting for a fixed number of rounds.

Algorithm 2.

Initialize $V := I_q$. At the start of each block:

1. Define

$$U_V(x) := \min_{y \in D} [r(x, y) + \lambda p_V(x, y)],$$

and choose an arm $x \in \operatorname{argmax}_{x' \in X} U_V(x')$.

2. Play x repeatedly. After n observations in the block, let

$$\bar{y}_n := \frac{1}{n} \sum_{i=1}^n y_i, \quad \bar{w}_n := w(x, \bar{y}_n).$$

End the block the first time

$$np_V(x, \bar{y}_n) \geq 1.$$

3. Update $V \leftarrow V + n\bar{w}_n\bar{w}_n^T$ and start a new block. At the horizon, stop even if the current block has not ended.
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Theorem 3 (Informal)

For fixed problem parameters, Algorithm 2, with $\lambda = \lceil \sqrt{T} \rceil$, has expected regret

$$R_T \leq \tilde{O}(\sqrt{T}).$$

The learner does not need to know S .

Proof sketch As before, each completed block at least doubles the determinant of V , so there are only $O(\log T)$ blocks.

By Lemma 1, with high probability, a block of length n has regret at most $\lambda np_V(x, \bar{y}) + \tilde{O}(n/\lambda)$. The stopping rule keeps $np_V(x, \bar{y})$ bounded by a constant, so the first term costs $O(\lambda)$ per block, regardless of its length. The remaining term costs about $1/\lambda$ per round. Summing both costs gives regret at most about $\lambda \log T + T/\lambda$. Ignoring logarithmic factors, choosing $\lambda = \lceil \sqrt{T} \rceil$ balances the two terms. Taking the concentration failure probability of order T^{-2} gives the same $\tilde{O}(\sqrt{T})$ bound in expectation. ■

The precise regret bound and detailed proof are in Appendix D.

3.1. Computational tractability

To implement the learner, we allow an additive error $\varepsilon > 0$ when maximizing U_V . This adds at most $n\varepsilon$ to the regret of a block of length n , and at most $T\varepsilon$ over all rounds. Taking $\varepsilon = (T+1)^{-2}$ therefore preserves the $\tilde{O}(\sqrt{T})$ regret bound.

Dualizing the penalty turns the planning step into a quadratic program with a fixed number of quadratic constraints. The weak-optimization result in Appendix A and sufficiently fine rounding give an implementation under the weak finite-precision model that runs in time polynomial in $|\mathcal{M}|$ and T and retains the same regret rate. The details are in Appendix D.

4. NP-hardness

Proofs are in Appendix E.

4.1. $D = \text{simplex}, X = \text{ball}$

Theorem 4

Consider a variation of the linear imprecise bandits problem where D is allowed to be a simplex. If, for some fixed $\beta > 0$, a randomized polynomial-time learner guarantees $E[R_T] \leq P(|\mathcal{M}|) T^{1-\beta}$ for every instance $\mathcal{M}$, then $\text{NP} \subseteq \text{BPP}$. If a deterministic learner achieves the same regret guarantee, then $\text{P} = \text{NP}$.

4.2. A Euclidean outcome ball and a polytope of arms

Theorem 5

Suppose we allow X to be a polytope in the linear imprecise bandits problem. If, for some fixed $\beta > 0$ and polynomial P , a randomized polynomial-time learner guarantees $R_T \leq P(|\mathcal{M}|) T^{1-\beta}$ for every instance $\mathcal{M}$, then $\text{NP} \subseteq \text{BPP}$. If a deterministic learner achieves the same regret guarantee, then $\text{P} = \text{NP}$.

4.3. Trilinear constraints with Euclidean arm and outcome balls

Theorem 6 *For the instances below, $F : \mathbb{R}^{d_X} \times \mathbb{R}^{d_Z} \times Y \rightarrow W$ is linear in each argument separately, where Y is the ambient outcome space and W is the constraint space. Write $F_{x,z}(y) := F(x, z, y)$, $L_z(x) := \ker F_{x,z}$, and $K_z(x) := L_z(x) \cap D$, where D lies in the hyperplane in which an added outcome coordinate equals one.*

There is a polynomial-time reduction from CLIQUE to imprecise-bandit instances with rational coefficients for which X is a full-dimensional Euclidean ball, D is a Euclidean ball in its affine hull, $Z = [1, 2]$ is a one-dimensional Euclidean ball, the reward is linear, and F is trilinear. Every set $K_z(x)$ is a singleton independent of z . The non-tangency inequality holds uniformly for every point in $L_z(x)$ outside D , with a constant $S > 0.88$.

For these instances, approximating V_z to inverse-polynomial additive accuracy is NP-hard even when z is known. Thus, if, for some fixed $\beta > 0$, a randomized polynomial-time learner guaranteed

$$R_T \leq P(|\mathcal{M}|) T^{1-\beta}$$

on every such instance for a polynomial P , then $\text{NP} \subseteq \text{BPP}$. For a deterministic learner, the same conclusion would be $\text{P} = \text{NP}$.

4.4. Computing IUCB's first arm is NP-hard even for Euclidean balls

Theorem 7 Write $V_{\text{IUCB}} := \max_{z \in Z, x \in X} v_z(x)$ for IUCB's optimistic value on its first round. A globally optimal first arm maximizes $\max_{z \in Z} v_z(x)$ over $x \in X$.

Computing V_{IUCB} exactly or finding a globally optimal first arm of IUCB is NP-hard, even when X , D , and Z are Euclidean balls and planning for each fixed hypothesis takes polynomial time.

4.5. Efficient planning does not imply efficient learning

Theorem 8

There is a family of rational imprecise-bandit instances indexed by integers $2 \leq k \leq n$, with descriptions of length polynomial in n . Each instance has finite arm and hypothesis sets, an outcome set that is a rational polytope, constraints $F = F_0 + F_1$ with $F_0 = 0$ and F_1 bilinear in the outcome and hypothesis, and a reward in $[0, 1]$ that is linear in the arm and in the outcome separately. An exact planner takes $O(n^2)$ time when the hypothesis is known.

Suppose that, for some fixed $\beta > 0$ and polynomial P , one uniform randomized policy runs in time polynomial in the description length $|\mathcal{M}|$ and horizon N and has expected regret at most

$$P(|\mathcal{M}|) N^{1-\beta}$$

for every instance in this family, every true hypothesis, and every compatible nature policy. Then $\text{CLIQUE} \in \text{RP}$ and hence $\text{NP} = \text{RP}$. The same conclusion holds if the guarantee is required only against stationary deterministic nature policies. A deterministic learner with the same guarantee would imply $\text{P} = \text{NP}$.

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Appendix

A. Quadratically constrained quadratic programming

For symmetric matrices $Q_i \in \mathbb{R}^{d \times d}$, a **quadratically constrained quadratic program** (QCQP) in $z \in \mathbb{R}^d$ has the form

$$\min_{z \in \mathbb{R}^d} g_0(z) \quad \text{subject to} \quad g_i(z) \leq 0, \quad i = 1, \dots, m,$$

where $g_i(z) := z^T Q_i z + 2p_i^T z + r_i$; for complexity statements, all entries are rational. A quadratic equality $h(z) = 0$ is exactly the pair $h(z) \leq 0$ and $-h(z) \leq 0$ [3].

QCQP describes algebraic form, whereas convex programming describes geometry. The displayed minimization problem is a convex program when every $Q_i, i = 0, \dots, m$, is positive semidefinite and all equalities are affine [7]. Without those restrictions it can be nonconvex: general QCQP is NP-hard even when only the objective is quadratic and its Hessian has one negative eigenvalue [8].

The polynomial-time conclusion for our nonconvex instance has two steps.

1. First, fix the number m of constraints. Bienstock's theorem states that, for rational quadratics $g_0, \dots, g_m$, if at least one constraint $g_i(z) \leq 0$ has a positive-definite quadratic part, then for every $\varepsilon \in (0, 1)$ an algorithm runs in time $\text{poly}(L, \log(\frac{1}{\varepsilon}))$, where L is the input bit length, and either proves infeasibility or returns $\hat{z}$ such that

$$g_i(\hat{z}) \leq \varepsilon, \quad i = 1, \dots, m,$$

and $g_0(\hat{z}) \leq g_0(z) + \varepsilon$ for every feasible z [3]. The underlying weak-feasibility step reduces to a fixed number of homogeneous quadratic equations on a sphere, handled by Barvinok's method; binary search then gives the objective guarantee [3], [9].

2. Second, the planning reduction in Appendix D has six quadratic inequalities. The redundant ellipsoid constraint on (x, z, s) has a positive-definite quadratic part, so the QCQP satisfies Bienstock's hypotheses. Applying the theorem to the negative objective computes the planning maximum to weak additive accuracy ε in time $\text{poly}(L, \log(\frac{1}{\varepsilon}))$. The same appendix explains how to repair the solver's output to obtain a feasible arm. If either radius is zero, the problem first reduces to a lower-dimensional instance.

B. MAX-CUT

Let $G = (V, E)$ be a finite undirected simple graph. For $S \subseteq V$, the edge boundary of S is

$$\delta_G(S) := \{\{u, v\} \in E : |\{u, v\} \cap S| = 1\}.$$

The decision problem MAX-CUT takes as input G and an integer $k \in \{0, \dots, |E|\}$, and asks whether

$$\exists S \subseteq V \quad \text{such that} \quad |\delta_G(S)| \geq k.$$

A set S satisfying this inequality is a polynomial-size certificate, so MAX-CUT belongs to NP. Karp proved that weighted MAX-CUT is NP-complete [10]. Garey, Johnson, and Stockmeyer subsequently proved that the unweighted problem defined above remains NP-complete even when G is cubic [11].

The associated optimization problem computes

$$M_G := \max_{S \subseteq V} |\delta_G(S)|.$$

Since the decision problem asks whether $M_G \geq k$, computing M_G is NP-hard.

C. Regret analysis of the $T^{\frac{2}{3}}$ learner

We give the precise statement and proof of Theorem 2. We use the coordinates and notation from Section 2, and assume that the planning steps in Algorithm 1 are solved exactly. A full block is *informative* if it causes an update, and *uninformative* otherwise.

For the analysis, fix $\delta \in (0, 1)$ and define

$$\begin{aligned} J_{\max} &:= \left\lceil q \log_2 \left(1 + 3 \frac{T}{q} \right) \right\rceil, \\ h_T^2 &:= 8d_D \log \left(\frac{2d_D T^2}{\delta} \right), \\ B_T^2 &:= 1 + J_{\max} h_T^2, \\ G &:= \sqrt{3} (1 + S^{-1}) \|b\|_2, \\ C_r &:= \max_{X \times D} r - \min_{X \times D} r. \end{aligned}$$

We write J for the number of updates actually made. We will show that $J \leq J_{\max}$.

If $w(x, y)$ is close to $\mathcal{L}$, then $r(x, y)$ cannot be much smaller than the worst feasible reward $v^*(x)$. We first prove this.

Lemma 2

For every $x \in X$ and $y \in D$,

$$v^*(x) \leq r(x, y) + G \operatorname{dist}(w(x, y), \mathcal{L}).$$

Proof Let $N := \ker C_{z^*}$ and let Q be the orthogonal projector onto $N^\perp$. Every arm has a feasible outcome, so we can write the minimum-norm solution of the affine constraints as $f(x) = Ax + e \in N^\perp$. Thus

$$K^*(x) = (Ax + e + N) \cap D.$$

Since there is a feasible outcome in the unit ball for every arm, $\|Ax + e\|_2 \leq 1$ on X . Taking $x = 0$ gives $\|e\|_2 \leq 1$, and comparing opposite unit vectors gives $\|A\|_{\text{op}} \leq 1$. Now define

$$\Phi(s, x, y) := Qy - Ax - se.$$

This map has operator norm at most $\sqrt{3}$ and kernel $\mathcal{L}$: the original constraints are equivalent to $Qy = Ax + e$, and homogenizing either equation gives the same subspace. Therefore

$$\text{dist}(y, Ax + e + N) = \|\Phi(1, x, y)\|_2 \leq \sqrt{3} \text{dist}(w(x, y), \mathcal{L}).$$

Let p be the point in $Ax + e + N$ closest to y . Since $y \in D$, the non-tangency condition gives

$$\text{dist}(p, K^*(x)) \leq S^{-1} \text{dist}(p, D) \leq S^{-1} \|p - y\|_2,$$

where the first inequality is immediate if $p \in D$, since then $p \in K^*(x)$. The triangle inequality now gives

$$\text{dist}(y, K^*(x)) \leq (1 + S^{-1}) \text{dist}(y, Ax + e + N).$$

Moving y to the nearest point in $K^*(x)$ changes the reward by at most $\|b\|_2$ times this distance, which proves the claim. $\blacksquare$

We also need to know how close a block average is to the average of the true conditional means.

Lemma 3

Let m_t be the conditional mean of y_t given the history and the chosen arm on round t . For each interval in the first T rounds, write n for its length and $\bar{y}$ and $\bar{m}$ for the averages of y_t and m_t over that interval. With probability at least $1 - \delta$, the bound

$$n \|\bar{y} - \bar{m}\|_2^2 \leq h_T^2 \tag{2}$$

holds simultaneously on all these intervals.

Proof Each coordinate of $y_t - m_t$ is a martingale difference bounded in absolute value by two. For any fixed interval of length n and coordinate i , Azuma–Hoeffding gives

$$\Pr \left(|\bar{y}_i - \bar{m}_i| > \frac{h_T}{\sqrt{nd_D}} \right) \leq 2 \exp \left(-\frac{h_T^2}{8d_D} \right) = \frac{\delta}{d_D T^2}.$$

There are d_D coordinates and at most T^2 intervals. A union bound and summing the squared coordinate bounds give the claim. $\blacksquare$

We need to know how far a point in E_V can be from $\mathcal{L}$. The next lemma bounds this distance using the noise bound we just proved. We can then apply Lemma 2 to the outcomes the learner treats as feasible.

Lemma 4

With probability at least $1 - \delta$, after any number J of updates in Algorithm 1, the matrix V satisfies, for every $w \in \mathbb{R}^q$,

$$\text{dist}(w, \mathcal{L}) \leq \sqrt{\frac{1 + Jh_T^2}{n}} \|w\|_{V^{-1}}.$$

Proof Number the informative blocks $j = 1, \dots, J$. Write x_j for the arm played in block j , and $\bar{y}_j$ and $\bar{m}_j$ for its average observation and average conditional mean. With $\bar{w}_j := w(x_j, \bar{y}_j)$, the update rule gives

$$V = n^{-1}I_q + \sum_{j=1}^J \bar{w}_j \bar{w}_j^T.$$

Fix $u \in \mathcal{L}^\perp$ with $\|u\|_2 \leq 1$. Since the arm is held fixed within each block, $w(x_j, \bar{m}_j) \in \mathcal{L}$. Each informative block has length n . By Lemma 3, with probability at least $1 - \delta$, the bound

$$(u^T \bar{w}_j)^2 = (u^T (0, 0, \bar{y}_j - \bar{m}_j))^2 \leq \frac{h_T^2}{n}$$

holds simultaneously for all informative blocks and all such u . On this event, $u^T V u \leq \frac{1+Jh_T^2}{n}$. For any w , Cauchy–Schwarz gives $u^T w \leq \sqrt{u^T V u} \|w\|_{V^{-1}}$. Taking the supremum over the unit ball of $\mathcal{L}^\perp$ proves the claim. $\blacksquare$

Theorem 9

Algorithm 1 satisfies, with probability at least $1 - \delta$,

$$TV^\star - \sum_{t=1}^T r(x_t, y_t) \leq C_r n (J_{\max} + 1) + \frac{GB_T T}{\sqrt{n}}.$$

Its expected regret is at most the same expression plus $C_r T \delta$. Taking $n = \left\lceil T^{\frac{2}{3}} \right\rceil$ and $\delta = (T+1)^{-2}$ give $R_T \leq \tilde{O}\left(T^{\frac{2}{3}}\right)$ for fixed problem parameters.

Proof First, there cannot be very many informative blocks. Each one more than doubles the determinant:

$$\det(V + \bar{w}\bar{w}^T) = \det(V) (1 + p_V(x, \bar{y})) > 2 \det(V).$$

If there are J updates, then $J \leq \frac{T}{n}$. Each update adds at most three to the trace, so $\text{tr } V \leq \frac{q}{n} + 3J \leq \frac{q+3T}{n}$. The initial determinant is n^{-q} . Using the arithmetic–geometric mean inequality on the eigenvalues, we get

$$2^J \leq \frac{\det V}{\det(n^{-1}I_q)} \leq \left(1 + 3\frac{T}{q}\right)^q, \quad J \leq J_{\max}.$$

Since $J \leq J_{\max}$, Lemma 4 gives, with probability at least $1 - \delta$, simultaneously for every matrix V that the learner maintains and every $(x, y) \in X \times D$,

$$\text{dist}(w(x, y), \mathcal{L}) \leq \left(\frac{B_T}{\sqrt{n}}\right) \sqrt{p_V(x, y)}. \quad (3)$$

On this event, if $y \in \hat{K}_V(x)$, then $p_V(x, y) \leq 1$, so Lemma 2 gives $r(x, y) \geq v^\star(x) - G\frac{B_T}{\sqrt{n}}$. Taking the minimum over $\hat{K}_V(x)$, whenever this set is nonempty, gives

$$H_V(x) \geq v^*(x) - G \frac{B_T}{\sqrt{n}}. \quad (4)$$

Consider a full block in which we do not update the ellipsoid. Its empirical mean belongs to $\hat{K}_V(x)$. Such a block could not have used the empty-set rule, since that would force the mean outside the ellipsoid. Thus all the sets $\hat{K}_V(x')$ were nonempty, and we chose x by maximizing H_V . It follows that

$$r(x, \bar{y}) \geq H_V(x) = \max_{x' \in X} H_V(x') \geq V^* - G \frac{B_T}{\sqrt{n}}.$$

Since the reward is affine, $r(x, \bar{y})$ is the average reward in the block. The block's regret is therefore at most $GB_T\sqrt{n}$. There are at most $\frac{T}{n}$ such blocks, for a total of $GB_T\frac{T}{\sqrt{n}}$. An informative block may cost as much as $C_r n$, but there are at most $J_{\max}$ of them. A final incomplete block also costs at most $C_r n$. Adding these bounds proves the high-probability claim. Outside this event, regret is still at most $C_r T$, so taking expectations adds at most $C_r T \delta$. $\blacksquare$

C.A. Finite-precision implementation

Proof We justify the implementation in Section 2.1. The matrix always satisfies

$$n^{-1}I_q \preceq V \preceq (n^{-1} + 3T)I_q,$$

since each update adds a positive semidefinite matrix of operator norm at most three. In particular, E_V contains the Euclidean ball of radius $\frac{1}{\sqrt{n}}$.

The distance formula follows by writing distance to $2E_V$ as

$$\text{dist}(w, 2E_V) = \max_{\|z\|_2 \leq 1} \left[z^T w - 2\sqrt{z^T V z} \right].$$

Minimizing over $w = (1, x, y)$ with $y \in D$ and exchanging the minimum and maximum gives the claimed expression for $d_V(x)$. The exchange is valid because the domains are compact and convex, and the objective is affine in y and concave in z .

Suppose the empty-set test has not returned an arm. Its additive guarantee gives $d_V(x) \leq 2\eta$ for every x . Thus, for every arm, there is a $y_0 \in D$ such that $w_0 := w(x, y_0)$ satisfies

$$\|w_0\|_{V^{-1}} \leq 2 + 2\eta\sqrt{n} \leq \frac{11}{5}.$$

This point lies strictly inside $3E_V$. We may therefore apply convex duality to the constraint $w(x, y) \in 3E_V$, whose indicator has the representation

$$\iota_{3E_V}(w) = \sup_z \left[z^T w - 3\sqrt{z^T V z} \right],$$

where the indicator is zero on the set and $+\infty$ outside it. Strong duality holds because $y_0 \in D$ and w_0 is in the interior of $3E_V$ [7]. Minimizing the resulting affine expression over the unit ball D gives the formula for H_V^{out} , initially with a supremum over all z .

To bound z , write the expression inside that supremum as $F_x(z)$. Using $\|y_0\|_2 \leq 1$ and Cauchy-Schwarz in the V norm gives

$$\begin{aligned}
F_x(z) &\leq b^T y_0 + z^T w_0 - 3\sqrt{z^T V z} \\
&\leq \|b\|_2 - \frac{4}{5}\sqrt{z^T V z} \\
&\leq \|b\|_2 - \frac{4}{5\sqrt{n}}\|z\|_2.
\end{aligned}$$

Since $F_x(0) = -\|b\|_2$, the supremum is attained at a point with $\|z\|_2 \leq \left(\frac{5}{2}\right)\sqrt{n}\|b\|_2$. This is smaller than the bound M used in the implementation.

We next give the quadratic programs explicitly. For the distance problem take $(\alpha, g, R) = (2, 0, 1)$, and for the planning problem take $(\alpha, g, R) = (3, b, M)$. Replace $\alpha\sqrt{z^T V z} + \|g + z_D\|_2$ by $\alpha t + s$ and impose

$$\|x\|_2^2 \leq 1, \quad \|z\|_2^2 \leq R^2, \quad z^T V z \leq t^2, \quad \|g + z_D\|_2^2 \leq s^2,$$

$$0 \leq t \leq 1 + (1 + 3T)R, \quad 0 \leq s \leq 1 + \|g\|_1 + R.$$

The remaining objective is $z_0 + z_X^T x - \alpha t - s$, with the additional term $a^T x + c$ for planning. The upper bounds leave room beyond the largest possible values of the two norms. Adding a redundant ball containing all these bounded variables gives nine quadratic inequalities, one with a positive-definite quadratic part. The result in Appendix A therefore applies.

A weak solution may slightly violate these constraints. Project its x and z components onto their balls and round inward to rational grids. Replace t and s by rational upper approximations to the corresponding norms. The resulting objective is a valid lower bound on $d_V(x)$ or $H_V^{\text{out}}(x)$ at the returned arm. If the weak constraint error is τ , projection moves each vector by at most $\sqrt{\tau}$; the changes in the two norms and the quadratic objective are bounded by a polynomial in the numerical data and T times $\sqrt{\tau}$. Thus both the required objective accuracy and a certified lower bound can be obtained with polynomially many precision bits. For the distance test, include all these errors in its additive budget η : a returned lower bound above η then certifies emptiness, and a lower bound at most η implies $\max_{x \in X} d_V(x) \leq 2\eta$.

Finally, round each observed outcome inward to a common rational grid, with Euclidean error at most $\tau_y := \frac{(T+1)^{-4}}{1+\|b\|_1}$. Use these rounded observations in the block averages and updates. The interval bound in Lemma 3 remains valid with h_T^2 replaced by $2h_T^2 + 2T\tau_y^2$. The matrix-distance proof and the determinant argument therefore still apply. An uninformative rounded mean belongs to E_V , and hence to both the inner and outer sets. Consequently such a block must have used the planning rule. Applying Lemma 2 and the matrix-distance bound on $3E_V$ gives the same regret estimate with the geometric term multiplied by three, an additional $T\epsilon$ for planning error, and at most $\|b\|_2 T\tau_y$ for replacing the actual rewards by rounded ones.

The common grids keep all stored matrices rational with polynomial bit length. Matrix inversion and the comparison $p_V > 1$ can then be performed exactly in polynomial time. There are at most $\lceil \frac{T}{n} \rceil$ planning calls, each taking time polynomial in the input bit length, T , and $\log\left(\frac{1}{\epsilon}\right)$. This proves the claimed implementation and regret bound. ■

D. Regret analysis of the $\sqrt{T}$ learner

We give the precise statement and proof of Theorem 3. We use the coordinates and notation from Section 3, including $\lambda = \lceil \sqrt{T} \rceil$.

For the analysis, fix $\delta \in (0, 1)$ and use the quantities $J_{\max}, h_T, B_T, G, C_r$ defined in Appendix C. We use the concentration bound from Lemma 3 and adapt the proof of Lemma 4 to the new matrix update.

Theorem 10

Consider the variant of Algorithm 2 in which each planning call maximizes U_V to within $\varepsilon \geq 0$. Run in exact arithmetic, it satisfies with probability at least $1 - \delta$,

$$TV^* - \sum_{t=1}^T r(x_t, y_t) \leq 4\lambda (J_{\max} + 1) + \frac{G^2 B_T^2 T}{4\lambda} + T\varepsilon,$$

With $\varepsilon = \delta = (T + 1)^{-2}$, we can implement this variant in time polynomial in $|\mathcal{M}|$ and T under the weak finite-precision model, with expected regret

$$R_T \leq \tilde{O} \left(\left[q + (1 + S^{-1})^2 \|b\|_2^2 q d_D \right] \sqrt{T} \right) + O \left(\frac{C_r}{T} \right).$$

The learner does not need to know S or the analysis quantities $J_{\max}, h_T, B_T, G$.

Proof As before, each completed block at least doubles the determinant. The stopping rule gives

$$\det(V + n\bar{w}\bar{w}^T) = \det(V) (1 + np_V(x, \bar{y})) \geq 2 \det(V).$$

Since $\|\bar{w}\|_2^2 \leq 3$, the final trace is at most $q + 3T$. Thus, if we complete J blocks, $2^J \leq \det V \leq \left(1 + 3\frac{T}{q}\right)^q$, so $J \leq J_{\max}$.

We also need to bound how far $np_V(x, \bar{y}_n)$ can jump above one on the last observation. For $n \geq 2$, convexity gives

$$np_V(x, \bar{y}_n) \leq (n-1)p_V(x, \bar{y}_{n-1}) + p_V(x, y_n) < 1 + 3 = 4.$$

The first term is less than one because we did not stop at $n-1$. The second is at most three because $V \succeq I_q$ and $p_V(x, y_n) \leq \|w(x, y_n)\|_2^2 \leq 3$. If we stop at $n=1$, the bound is at most three. If the horizon ends before the test succeeds, it is less than one. Thus every block satisfies

$$np_V(x, \bar{y}) \leq 4. \tag{5}$$

We now adapt the proof of Lemma 4. Number the completed blocks $j = 1, \dots, J$, and let n_j be the length of block j . Write x_j for its arm, $\bar{y}_j$ and $\bar{m}_j$ for its average observation and conditional mean, and $\bar{w}_j := w(x_j, \bar{y}_j)$. The matrix now has the form

$$V = I_q + \sum_{j=1}^J n_j \bar{w}_j \bar{w}_j^T.$$

By Lemma 3, with probability at least $1 - \delta$, the bound $(u^T \bar{w}_j)^2 \leq \frac{h_T^2}{n_j}$ holds simultaneously for all completed blocks and all $u \in \mathcal{L}^\perp$ with $\|u\|_2 \leq 1$. On this event,

$$u^T V u \leq 1 + \sum_{j=1}^J n_j \left(\frac{h_T^2}{n_j} \right) = 1 + J h_T^2 \leq B_T^2.$$

This is where the weight n_j matters: it cancels the $\frac{1}{n_j}$ in the squared noise bound. Each block contributes at most h_T^2 , however long it is. Applying Cauchy–Schwarz and taking the supremum over these u gives

$$\text{dist}(w, \mathcal{L}) \leq B_T \|w\|_{V^{-1}}. \quad (6)$$

Next, we compare the score with the true value of an arm. Lemma 2 gives, for every $x \in X$ and $y \in D$,

$$v^*(x) \leq r(x, y) + G B_T \sqrt{p_V(x, y)}.$$

The error bound grows with $\sqrt{p_V(x, y)}$, while the penalty grows with $p_V(x, y)$. Writing $u := \sqrt{p_V(x, y)}$ and completing the square,

$$r(x, y) + \lambda p_V(x, y) \geq v^*(x) + \lambda u^2 - G B_T u \geq v^*(x) - \frac{G^2 B_T^2}{4\lambda}.$$

Taking the minimum over $y \in D$, we see that the score underestimates the true value by at most ζ_T :

$$U_V(x) \geq v^*(x) - \zeta_T, \quad \zeta_T := \frac{G^2 B_T^2}{4\lambda}. \quad (7)$$

We choose an arm whose score is within ε of the highest score, so $U_V(x) \geq V^* - \zeta_T - \varepsilon$. The block's empirical mean is in D , and the reward is affine. Its regret is therefore at most

$$n(V^* - r(x, \bar{y})) \leq \lambda n p_V(x, \bar{y}) + n(\zeta_T + \varepsilon) \leq 4\lambda + n(\zeta_T + \varepsilon).$$

Adding this over at most $J_{\max} + 1$ blocks proves the high-probability bound. If the noise bound fails, regret is still at most $C_r T$, so taking expectations adds at most $C_r T \delta$.

The difference from the warmup is that we now pay at most 4λ in penalty per block, no matter how long the block lasts. We also pay for the score underestimating the true value, but this costs only ζ_T per round, which decreases as $\frac{1}{\lambda}$. The two terms to balance are therefore $\lambda J_{\max}$ and $\frac{T}{\lambda}$, giving the square-root rate.

We still need to implement the planning step. First, rewrite the penalty as

$$\lambda w^T V^{-1} w = \max_z \left[z^T w - \frac{z^T V z}{4\lambda} \right].$$

The maximizer is $z = 2\lambda V^{-1} w$. Since $V \succeq I_q$ and $\|w\|_2 \leq \sqrt{3}$, we can restrict the maximum to the ball $\|z\|_2 \leq 4\lambda$ for every $x \in X$ and $y \in D$. For fixed x , the expression is affine in y and concave in z , and both domains are compact and convex. We can therefore exchange the minimum and maximum. Writing $z = (z_0, z_X, z_D)$ gives

$$U_V(x) = a^T x + c + \max_{\|z\|_2 \leq 4\lambda} \left[z_0 + z_X^T x - \|b + z_D\|_2 - \frac{z^T V z}{4\lambda} \right]. \quad (8)$$

We can now maximize U_V by solving the QCQP

$$\max_{x, z, s} \quad a^T x + c + z_0 + z_X^T x - s - \frac{z^T V z}{4\lambda}$$

subject to

$$\|x\|_2^2 \leq 1, \quad \|z\|_2^2 \leq 16\lambda^2, \quad \|b + z_D\|_2^2 \leq s^2, \quad 0 \leq s \leq H,$$

where $H := 1 + \|b\|_1 + 4\lambda$ is rational. We also add the redundant constraint

$$\|x\|_2^2 + \frac{\|z\|_2^2}{16\lambda^2} + \frac{s^2}{H^2} \leq 3.$$

This gives six quadratic inequalities. The last has a positive-definite quadratic part in all variables, so we can apply Bienstock's weak-optimization theorem [3] as described in Section A.

The solver may return a point that slightly violates the constraints. We can make its output $(\hat{x}, \hat{z}, \hat{s})$ feasible by projecting $\hat{x}$ and $\hat{z}$ onto their respective balls. Call the resulting points x' and z' , and put $s' := \|b + z'_D\|_2$. If each constraint violation is at most $\tau \leq 1$, then

$$\|x' - \hat{x}\|_2 \leq \sqrt{\tau}, \quad \|z' - \hat{z}\|_2 \leq \sqrt{\tau}, \quad s' - \hat{s} \leq 4\sqrt{\tau}.$$

The last bound only controls an increase, but this is all we need: the objective has coefficient -1 on s . Since $\|V\|_{\text{op}} \leq 1 + 3T$, the repair decreases the objective by at most $K_T \sqrt{\tau}$, where we can take the rational bound

$$K_T := 20(1 + \|a\|_1 + \lambda + T).$$

Choose $\tau \leq \min \left(\frac{1}{4}, \left(\frac{\varepsilon}{4(K_T+1)} \right)^2 \right)$. Including the solver's objective error, the objective at the repaired point is then within $\frac{\varepsilon}{2}$ of the optimum. By Equation 8, the same lower bound holds for $U_V(x')$. The number of precision bits we need is polynomial in $|\mathcal{M}|$, $\log(T+1)$, and $\log(\frac{1}{\varepsilon})$, even when the reward coefficients are large.

Finally, we round the played arms and observed outcomes to common rational grids inside their unit balls. Truncating each coordinate toward zero keeps these points in the balls. The function U_V is Lipschitz in x with constant at most $\|a\|_2 + 2\sqrt{3}\lambda$, so we choose the arm grid finely enough that rounding costs at most $\frac{\varepsilon}{2}$. For outcomes, keep the Euclidean error at most $\tau_y := (T+1)^{-4}$ and use the rounded observations in the block averages and updates. Their interval averages still satisfy Equation 2 after replacing h_T^2 by $2h_T^2 + 2T\tau_y^2$. The bounds on the determinant and the penalty per block stay the same. Replacing actual rewards by rewards at the rounded observations changes the cumulative reward by at most $\|b\|_2 T \tau_y$. We therefore obtain the same expected-regret rate, with universal changes to the displayed high-probability constants.

The stored matrices are rational, with eigenvalues between 1 and $1 + 3T$. The common grids keep bit lengths polynomial when we form block averages, update and invert the matrices, and compare the stopping quantity with its threshold exactly. There are at most $J_{\max} + 1$ planning

calls, and the QCQP reduction above makes each call run in polynomial time. This proves the computational claim. $\blacksquare$

E. NP-hardness proofs

E.A. Planning from learning

In this section we show that several natural variations of the linear imprecise bandits problem are NP-hard. Thus in some sense, the problem is the hardest problem that's still tractable.

We first show that efficient planning can be reduced to efficient learning. Most proofs below show that planning is already NP-hard. Due to the reduction from planning to learning, this shows that learning is also NP-hard.

Lemma 5

Fix a linear imprecise bandit instance $\mathcal{M}$. Suppose that, for every hypothesis $z \in Z$, every arm x , and every $\eta \in (0, 1)$, one can compute an outcome $y_{z,\eta}(x)$ satisfying

$y_{z,\eta}(x) \in K_z(x)$ and $0 \leq r(x, y_{z,\eta}(x)) - v_z(x) \leq \eta$ in time polynomial in $|\mathcal{M}|$ and $\frac{1}{\eta}$. Suppose also that, for some fixed $\beta > 0$ and polynomial P , a polynomial-time learner guarantees $R_T \leq P(|\mathcal{M}|) T^{1-\beta}$

Then, for every $z \in Z$ and every $\varepsilon \in (0, 1)$, one can compute in time $\text{poly}(|\mathcal{M}|, \frac{1}{\varepsilon})$ an arm $\hat{x}$ such that with probability at least $2/3$, $|V_z - v_z(\hat{x})| \leq \varepsilon$.

Proof Fix $z \in Z$ and $\varepsilon \in (0, 1)$, set $\eta := \frac{\varepsilon}{6}$, and simulate the learner on $\mathcal{M}$ for

$$T := \left\lceil \max \left(1, \left(6 \frac{P(|\mathcal{M}|)}{\varepsilon} \right)^{\frac{1}{\beta}} \right) \right\rceil$$

rounds. Whenever the learner plays x_t , let nature use the point mass at $y_{z,\eta}(x_t)$. This is allowed because $y_{z,\eta}(x_t) \in K_z(x_t)$. Record the rewards and choose the round with the largest reward:

$$\tilde{v}_t := r(x_t, y_{z,\eta}(x_t)), \quad \hat{t} \in \operatorname{argmax}_{t=1,\dots,T} \tilde{v}_t,$$

and return $\hat{x} := x_{\hat{t}}$.

The approximation guarantee gives $v_z(\hat{x}) \geq \tilde{v}_{\hat{t}} - \eta$. Our choice of $\hat{t}$ therefore gives

$$0 \leq V_z - v_z(\hat{x}) \leq V_z - \tilde{v}_{\hat{t}} + \eta \leq V_z - \frac{1}{T} \sum_{t=1}^T \tilde{v}_t + \eta.$$

Taking expectations and using the definition of R_T for this nature policy,

$$\mathbb{E}[V_z - v_z(\hat{x})] \leq \frac{R_T}{T} + \eta \leq P(|\mathcal{M}|) T^{-\beta} + \eta \leq \frac{\varepsilon}{3}.$$

The gap $V_z - v_z(\hat{x})$ is nonnegative, so Markov's inequality shows that it exceeds ε with probability at most $\frac{1}{3}$. The horizon and the simulation time are polynomial in $|\mathcal{M}|$ and $\frac{1}{\varepsilon}$ because β is fixed. $\blacksquare$

Below we show that the “smoothness” of X and D are crucial for efficiency. It is easy to land in the NP-hard territory if we make either X or D a convex shape with edges and corners.

E.B. Proof of Theorem 4

First we show that if, instead of a Euclidean ball, we allow D to be a simplex, then even known hypothesis planning becomes NP-hard. Then, using the planning to learning reduction (Lemma 5), we conclude that learning is NP-hard.

Proof We reduce from MAX-CUT, as defined in Appendix B. Let (G, k) be an instance, where $G = (V, E)$ has n vertices and $m \geq 1$ edges and $k \in \{1, \dots, m\}$. Orient the edges arbitrarily and let $B_G \in \mathbb{R}^{m \times n}$ be the resulting edge-vertex incidence matrix, i.e., if $e = \{i, j\}$ is oriented from j to i , then row e has 1 in column i , -1 in column j , and zeros elsewhere. Put

$$X := \{x \in \mathbb{R}^m : \|x\|_2 \leq 1\}, \quad Z := [1, 2].$$

Write an outcome as $y = (p, q, s) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$ and take

$$D := \left\{ (p, q, s) : p_i \geq 0, q_i \geq 0 \text{ for } i = 1, \dots, n, s \geq 0, \sum_{i=1}^n (p_i + q_i) + s = 1 \right\}.$$

Let $\gamma > 0$ be a constant to be chosen later. For every $z \in Z$, let

$$K_z(x) := \{(p, q, s) \in D : p - q = \gamma B_G^T x\}.$$

This is of the form in Equation 1, where

$$C_z(p, q, s) := p - q, \quad B_z x := -\gamma B_G^T x, \quad d_z := 0.$$

Let the reward be

$$r(x, (p, q, s)) := \sum_{i=1}^n (p_i + q_i).$$

The maps $z \mapsto B_z$, $z \mapsto C_z$, and $z \mapsto d_z$ are constant, hence affine, as required, and the reward lies in $[0, 1]$ on D .

In particular, $K_z(x)$ is independent of z .

Next we show that, for every $z \in Z$, (G, k) is a yes-instance of MAX-CUT if and only if the constructed imprecise bandit instance has an arm $x \in X$ satisfying

$$v_z(x) \geq 2\gamma\sqrt{k}.$$

A vector $\sigma \in \{-1, 1\}^n$ represents a partition of the vertices of G into the two sides $\{i : \sigma_i = 1\}$ and $\{i : \sigma_i = -1\}$. Let $c_G(\sigma)$ denote the number of edges between these sides.

Lemma 6

For every $\sigma \in \{-1, 1\}^n$,

$$\|B_G \sigma\|_2 = 2\sqrt{c_G(\sigma)}.$$

Proof Fix $\sigma \in \{-1, 1\}^n$. For every edge $e = \{i, j\}$,

$$(B_G \sigma)_e^2 = (\sigma_i - \sigma_j)^2 = \begin{cases} 4 & \text{if } \sigma_i \neq \sigma_j \\ 0 & \text{otherwise} \end{cases}.$$

Summing over $e \in E$ and taking the square root proves the identity. $\blacksquare$

For the forward implication, suppose (G, k) is a yes-instance. Choose σ such that $c_G(\sigma) \geq k$, and define the arm

$$x_\sigma := B_G \frac{\sigma}{\|B_G \sigma\|_2}.$$

Since $c_G(\sigma) \geq k \geq 1$, Lemma 6 shows that x_σ is well-defined. Moreover, we get the following inequality.

Lemma 7

For every $\sigma \in \{-1, 1\}^n$ such that $B_G \sigma \neq 0$, we have $\|B_G^T x_\sigma\|_1 \geq \|B_G \sigma\|_2$. Moreover, equality holds when σ maximizes $\|B_G \sigma\|_2$ over $\{-1, 1\}^n$; in particular, equality holds for at least one σ .

Proof Choosing the sign of each coordinate gives

$$\|B_G^T x_\sigma\|_1 = \max_{\tau \in \{-1, 1\}^n} \tau^T B_G^T x_\sigma = \max_{\tau \in \{-1, 1\}^n} x_\sigma^T B_G \tau.$$

Taking $\tau = \sigma$ yields $\|B_G^T x_\sigma\|_1 \geq x_\sigma^T B_G \sigma = \|B_G \sigma\|_2$. Now let σ^* maximize $\|B_G \sigma\|_2$. Since $m \geq 1$, this maximum is positive, so x_{σ^*} is well-defined. For every $\tau \in \{-1, 1\}^n$, Cauchy–Schwarz gives

$$x_{\sigma^*}^T B_G \tau \leq \|x_{\sigma^*}\|_2 \|B_G \tau\|_2 \leq \|B_G \sigma^*\|_2.$$

Taking the maximum over τ gives the reverse inequality, proving equality. $\blacksquare$

The lemma above gives us a way to link properties of cuts with properties of arms. The missing ingredient is to show that the value of an arm has something to do with $\|B_G^T x\|_1$. We show this in the next two lemmas.

By construction, $x_\sigma \in X$. Since $K_z(x)$ is independent of z , write $K(x) := K_z(x) = \{(p, q, s) \in D : p - q = \gamma B_G^T x\}$.

Lemma 8

For every $x \in X$, the vector $t := \gamma B_G^T x$ satisfies $\|t\|_1 \leq \gamma \sqrt{2mn}$. Therefore, if $\gamma \leq \frac{1}{\sqrt{2mn}}$, then $\|t\|_1 \leq 1$.

Proof Each edge coordinate x_e contributes x_e and $-x_e$ to two coordinates of $B_G^T x$ before contributions from different edges are added. The triangle inequality therefore gives

$$\|t\|_1 = \gamma \|B_G^T x\|_1 \leq 2\gamma \|x\|_1 \leq 2\gamma \sqrt{m} \|x\|_2 \leq 2\gamma \sqrt{m} \leq \gamma \sqrt{2mn},$$

where the second inequality is Cauchy–Schwarz, the third uses $\|x\|_2 \leq 1$, and the last uses $n \geq 2$, which follows from $m \geq 1$. $\blacksquare$

Lemma 9

Suppose $\gamma \leq \frac{1}{\sqrt{2mn}}$. For each arm $x \in X$, set $t := \gamma B_G^T x$. Then $K(x)$ is nonempty, and for all z , $v_z(x) = \min_{y \in K(x)} r(x, y) = \|t\|_1 = \gamma \|B_G^T x\|_1$. Moreover, the minimum is attained at the outcome $y = (p, q, s)$ given by $p_i := \max(t_i, 0)$, $q_i := \max(-t_i, 0)$, and $s := 1 - \|t\|_1$.

Proof Fix $x \in X$ and set $t := \gamma B_G^T x$. The preceding lemma gives $\|t\|_1 \leq 1$. For the outcome in the statement, p, q, s are nonnegative, $p - q = t$, and

$$\sum_{i=1}^n (p_i + q_i) + s = \|t\|_1 + 1 - \|t\|_1 = 1.$$

Thus this outcome belongs to $K(x)$. For any $(p, q, s) \in K(x)$, nonnegativity gives $p_i + q_i \geq |p_i - q_i| = |t_i|$ for every i , and hence $r(x, (p, q, s)) \geq \|t\|_1$. The outcome in the statement attains equality in every coordinate, so its reward is the minimum. ■

Note that this also shows that given an arm x , we can compute $v_z(x)$ efficiently, which is one of the prerequisites for the application of Lemma 5.

For the final step, apply Lemma 9, Lemma 7, and Lemma 6 to obtain

$$v_z(x_\sigma) = \gamma \|B_G^T x_\sigma\|_1 \geq \gamma \|B_G \sigma\|_2 = 2\gamma \sqrt{c_G(\sigma)} \geq 2\gamma \sqrt{k}.$$

For the reverse implication, suppose there is an arm $x \in X$ satisfying $v_z(x) \geq 2\gamma \sqrt{k}$. By ℓ_1 - ℓ_∞ duality, choose $\sigma \in \{-1, 1\}^n$ such that

$$\|B_G^T x\|_1 = \sigma^T B_G^T x.$$

Using Lemma 9 and Lemma 6,

$$2\gamma \sqrt{k} \leq v_z(x) = \gamma \|B_G^T x\|_1 = \gamma x^T B_G \sigma \leq \gamma \|x\|_2 \|B_G \sigma\|_2 \leq 2\gamma \sqrt{c_G(\sigma)}.$$

Since $\gamma > 0$, this implies $c_G(\sigma) \geq k$, so (G, k) is a yes-instance. This proves the claimed equivalence.

To apply Lemma 5, we need to show that *approximate* planning is hard. Although the lemma returns only an approximate optimizer, this is sufficient because the maximum cut size M_G is an integer between 0 and m , and the corresponding planning values are $V_z = 2\gamma \sqrt{M_G}$. Taking $\gamma := \frac{1}{2mn}$ ensures that every $K_z(x)$ is nonempty and that consecutive planning values are separated by at least $\frac{1}{2m^2n}$. Thus an additive approximation with error $\varepsilon := \frac{1}{6m^2n}$ distinguishes $M_G \geq k$ from $M_G \leq k - 1$.

By Lemma 9, we can compute both the value of the returned arm and a minimizing outcome in polynomial time. We can therefore decide MAX-CUT by comparing the returned arm's value with the midpoint between $2\gamma \sqrt{k-1}$ and $2\gamma \sqrt{k}$, computed to sufficient precision. Hence Lemma 5 implies that a randomized polynomial-time learner would place MAX-CUT in BPP, and therefore that $\text{NP} \subseteq \text{BPP}$. A deterministic learner would instead imply $\text{P} = \text{NP}$. ■

E.C. Proof of Theorem 5

We now keep the Euclidean outcome ball and the additive bilinear constraint, but allow the arm set to be a polytope given by rational inequalities. Planning is NP-hard even when the hypothesis is known and the arm set is the cube. The cube has only $2n$ defining inequalities, but 2^n vertices.

Proof We again reduce from MAX-CUT. Let $G = (V, E)$ have n vertices and $m \geq 1$ edges. Orient the edges arbitrarily and let $B_G \in \mathbb{R}^{m \times n}$ be the edge-vertex incidence matrix, so that

$$(B_G x)_e = x_i - x_j$$

for an edge $e = \{i, j\}$. Put

$$\varepsilon := \frac{1}{4m}, \quad X := [-1, 1]^n, \quad Z := [1, 2].$$

Write outcomes as $y = (u, s) \in \mathbb{R}^m \times \mathbb{R}$ and take

$$D := \left\{ (u, s) : \|u\|_2^2 + s^2 \leq 1 \right\}.$$

With $W = \mathbb{R}^m$, define

$$F_0(x, z) := -\varepsilon z B_G x, \quad F_1((u, s), z) := zu,$$

The constraint is $F_0(x, z) + F_1((u, s), z) = 0$, and the reward is $r(x, (u, s)) := s$. Both constraint maps are bilinear and the reward is linear. Since $z \neq 0$, the constraint is equivalent to

$$u = \varepsilon B_G x,$$

so the feasible outcomes do not depend on z . Every $x \in X$ satisfies

$$\|B_G x\|_2^2 = \sum_{\{i, j\} \in E} (x_i - x_j)^2 \leq 4m,$$

so

$$\|\varepsilon B_G x\|_2^2 \leq \frac{1}{4m} \leq \frac{1}{4}.$$

Thus the feasible outcome set $K_z(x)$ is nonempty for every $x \in X$ and $z \in Z$. Minimizing s over this set gives

$$v_z(x) = -\sqrt{1 - \varepsilon^2 \|B_G x\|_2^2}.$$

Set $q(x) := \|B_G x\|_2^2$. With every coordinate except x_i fixed, $q(x)$ is a convex quadratic in x_i , so replacing x_i by one of the endpoints of $[-1, 1]$ does not decrease q . Applying this replacement one coordinate at a time produces a sign vector $\sigma \in \{-1, 1\}^n$ with $q(\sigma) \geq q(x)$. At a sign vector,

$$q(\sigma) = \sum_{\{i, j\} \in E} (\sigma_i - \sigma_j)^2 = 4 |\text{cut}(\sigma)|.$$

Consequently,

$$\max_{x \in [-1, 1]^n} q(x) = 4 \text{MaxCut}(G).$$

Since $v_z(x)$ is strictly increasing in $q(x)$, if the maximum cut has size k , the optimal robust value is

$$V_k = -\sqrt{1 - \frac{k}{4m^2}}.$$

For $k = 1, \dots, m$, consecutive possible values satisfy

$$V_k - V_{k-1} = \frac{\frac{1}{4m^2}}{\sqrt{1 - \frac{k-1}{4m^2}} + \sqrt{1 - \frac{k}{4m^2}}} \geq \frac{1}{8m^2}.$$

Hence an estimate of V_z within

$$\Delta := \frac{1}{32m^2}$$

identifies k after we compute the $m + 1$ candidate values to polynomially many bits. Thus approximating the optimal robust value is NP-hard.

We now show how to use the learner to find a maximum cut. To simulate the learner, we need rational outcomes in $K_z(x)$. The outcome attaining $v_z(x)$ need not be rational, so we compute a rational feasible approximation. Suppose the learner in the theorem exists and set

$$\eta := \frac{\Delta}{6}, \quad T := \left\lceil \left(6 \frac{P(|\mathcal{M}|)}{\Delta} \right)^{\frac{1}{\beta}} \right\rceil.$$

In the weak bit model the learner outputs rationally encoded arms. For each played arm x_t , we can use binary search to compute a rational a_t in polynomial time such that

$$0 \leq \sqrt{1 - \varepsilon^2 q(x_t)} - a_t \leq \eta, \quad a_t^2 \leq 1 - \varepsilon^2 q(x_t).$$

Let nature return the point

$$y_t := (\varepsilon B_G x_t, -a_t).$$

This is a rational point of $K_z(x_t)$, and its reward satisfies

$$0 \leq r(x_t, y_t) - v_z(x_t) \leq \eta.$$

Writing $V_z = V_k$ and

$$S_T := \sum_{t=1}^T (V_z - v_z(x_t)),$$

the definition of expected regret gives

$$E[S_T] \leq R_T + T\eta.$$

Applying the learner's regret guarantee,

$$E[S_T] \leq P(|\mathcal{M}|) T^{1-\beta} + T\eta.$$

If the learner never plays a Δ -optimal arm, then $S_T > \Delta T$. Markov's inequality and the choices of T and η imply

$$\Pr(\text{no } \Delta\text{-optimal arm is played}) \leq \frac{P(|\mathcal{M}|)}{\Delta T^\beta} + \frac{\eta}{\Delta} \leq \frac{1}{3}.$$

Select a played arm maximizing the rational quantity $q(x_t)$ and round its coordinates to end-points without decreasing q . With probability at least $\frac{2}{3}$, the resulting sign vector has robust value at least $V_k - \Delta$. A cut of size at most $k-1$ would instead have robust value at most $V_{k-1} \leq V_k - \frac{1}{8m^2} < V_k - \Delta$. The rounded sign vector therefore encodes a maximum cut. The horizon, the square-root approximations, and the rounding all have polynomial bit complexity, proving $\text{NP} \subseteq \text{BPP}$. If the learner is deterministic, the same argument succeeds with certainty and gives $\text{P} = \text{NP}$.

It remains to verify non-tangency. For a fixed x , put

$$a := \|\varepsilon B_G x\|_2, \quad \rho := \sqrt{1 - a^2} \geq \frac{\sqrt{3}}{2}.$$

The outcomes satisfying the affine constraint form the line $L_z(x)$, whose intersection with D is $K_z(x)$:

$$L_z(x) = \{(\varepsilon B_G x, s) : s \in \mathbb{R}\}, \quad K_z(x) = \{(\varepsilon B_G x, s) : |s| \leq \rho\}.$$

If $p = (\varepsilon B_G x, s) \in L_z(x)$ lies outside D and $t := |s| > \rho$, then

$$\text{dist}(p, K_z(x)) = t - \rho, \quad \text{dist}(p, D) = \sqrt{a^2 + t^2} - 1.$$

Factoring the second expression gives

$$\frac{\text{dist}(p, D)}{\text{dist}(p, K_z(x))} = \frac{t + \rho}{\sqrt{a^2 + t^2} + 1} \geq \rho \geq \frac{\sqrt{3}}{2}.$$

To check the first inequality, multiply by the positive denominator, subtract ρ from both sides, and square. This gives the equivalent inequality $(1 - \rho^2)(t^2 - \rho^2) \geq 0$. Thus the uniform non-tangency condition holds with $S = \frac{\sqrt{3}}{2}$. ■

E.D. Proof of Theorem 6

We keep the Euclidean arm and outcome balls, but allow the coefficients in the equations for y to depend on x . We add an outcome coordinate that is fixed to one, so that the constraints are linear in the full outcome vector with no constant term. Let Y be the ambient outcome space, let W be the constraint space, and let

$$F : \mathbb{R}^{d_X} \times \mathbb{R}^{d_Z} \times Y \rightarrow W$$

be linear in each argument separately. For each x and z , write $F_{x,z}(y) := F(x, z, y)$ and define

$$L_z(x) := \ker F_{x,z}, \quad K_z(x) := L_z(x) \cap D, \quad v_z(x) := \min_{y \in K_z(x)} r(x, y), \quad V_z := \max_{x \in X} v_z(x).$$

Here D is a Euclidean ball in the hyperplane where the added coordinate equals one. We will show that planning is now NP-hard even when z is known.

Let $G = (V, E)$ be an undirected graph with $V = \{1, \dots, n\}$ and $E = \{e_1, \dots, e_m\}$, where $m \geq 1$ and $e_\ell = \{i_\ell, j_\ell\}$. Put $d := n + m$. For $q = (u, w) \in \mathbb{R}^n \times \mathbb{R}^m$, define the homogeneous cubic

$$p_G(q) := \sum_{\ell=1}^m u_{i_\ell} u_{j_\ell} w_\ell.$$

Lemma 10

If $\omega(G)$ is the clique number of G , then

$$\max_{\|q\|_2=1} p_G(q)^2 = \frac{2}{27} \left(1 - \frac{1}{\omega(G)}\right).$$

Proof Write $\rho := \|u\|_2$ and $\sigma := \|w\|_2$. For fixed u and σ , Cauchy–Schwarz gives

$$\max_{\|w\|_2=\sigma} p_G(u, w) = \sigma \sqrt{\sum_{\ell=1}^m u_{i_\ell}^2 u_{j_\ell}^2}.$$

Write $u = \rho s$ with $\|s\|_2 = 1$ and set $\pi_i := s_i^2$. The Motzkin–Straus theorem [4] gives

$$\max_{\|s\|_2=1} \sum_{\{i,j\} \in E} s_i^2 s_j^2 = \max_{\pi_i \geq 0, \sum_i \pi_i = 1} \sum_{\{i,j\} \in E} \pi_i \pi_j = \frac{1}{2} \left(1 - \frac{1}{\omega(G)}\right).$$

Finally,

$$\max_{\rho^2 + \sigma^2 = 1, \rho \geq 0, \sigma \geq 0} \rho^2 \sigma = \frac{2}{3\sqrt{3}}.$$

Substituting these bounds into the first display and squaring proves the claim. Since p_G is odd, its maximum equals its maximum absolute value. ■

Proof of Theorem 6 We may restrict CLIQUE to instances with $m \geq 1$ and $3 \leq k \leq n$, since we can decide the other cases in polynomial time. Given (G, k) , use the cubic p_G above and write an arm as $x = (x_0, \xi) \in \mathbb{R} \times \mathbb{R}^d$. Take

$$X := \left\{ (x_0, \xi) : \left(x_0 - \frac{5}{3}\right)^2 + \|\xi\|_2^2 \leq 1 \right\}.$$

This is a full-dimensional Euclidean ball and $x_0 \geq \frac{2}{3}$ throughout X . The ratio $q := \frac{\xi}{x_0}$ ranges over exactly the ball $\|q\|_2 \leq \frac{3}{4}$. Indeed,

$$1 - \left(x_0 - \frac{5}{3}\right)^2 - \frac{9}{16} x_0^2 = -\frac{(15x_0 - 16)^2}{144} \leq 0,$$

so every ratio has norm at most $\frac{3}{4}$. Conversely, for any such q , the choice $x_0 = \frac{16}{15}$ and $\xi = x_0 q$ belongs to X .

Let $Z := [1, 2]$, and take the outcome and constraint spaces to be

$$Y := \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^{d \times d} \times \mathbb{R}^{d \times d \times d},$$

$$W := \mathbb{R}^d \times \mathbb{R}^{d \times d} \times \mathbb{R}^{d \times d \times d},$$

and write $y = (y_0, \eta, \Theta, \Xi) \in Y$. We fix $y_0 = 1$ and take the remaining coordinates to lie in the Euclidean unit ball:

$$D := \left\{ (1, \eta, \Theta, \Xi) : \|\eta\|_2^2 + \|\Theta\|_F^2 + \|\Xi\|_F^2 \leq 1 \right\}.$$

For $x = (x_0, \xi), z \in \mathbb{R}$, and $y \in Y$, define $F(x, z, y)$ coordinatewise by

$$F_i(x, z, y) := z \left(x_0 \eta_i - \frac{1}{2} \xi_i y_0 \right),$$

$$F_{ij}(x, z, y) := z (x_0 \Theta_{ij} - \xi_i \eta_j),$$

$$F_{ij\ell}(x, z, y) := z (x_0 \Xi_{ij\ell} - \xi_i \Theta_{j\ell}).$$

Each term contains one coordinate from each of x, z , and y , so F is trilinear. Define the reward by

$$r(x, y) := \frac{1}{m} \sum_{\ell=1}^m \Xi_{i_\ell, j_\ell, n+\ell}.$$

It is linear and independent of x ; its coefficient vector has Euclidean norm $\frac{1}{\sqrt{m}} \leq 1$.

Fix $x \in X$ and $z \in Z$. Since $zx_0 \neq 0$ and $y_0 = 1$ on D , we can solve $F(x, z, y) = 0$ first for η , then for Θ , and then for Ξ . This gives

$$\eta_i = \frac{1}{2} q_i, \quad \Theta_{ij} = \frac{1}{2} q_i q_j, \quad \Xi_{ij\ell} = \frac{1}{2} q_i q_j q_\ell.$$

Writing $t := \|q\|_2 \leq \frac{3}{4}$, the squared norm of (η, Θ, Ξ) is

$$\frac{1}{4} (t^2 + t^4 + t^6) \leq \frac{4329}{16384} < 1.$$

Thus this point lies in D , so $K_z(x)$ is a singleton independent of z . The map $F_{x,z} : Y \rightarrow W$ is also onto: for any right-hand side, set $y_0 = 0$ and solve successively for η, Θ , and Ξ .

Within the hyperplane $y_0 = 1$, the equations have just one solution, and that solution lies in D . So there are no solutions outside D to check in the non-tangency condition. We can also prove the distance inequality for points in the full kernel $L_z(x)$. Since $F_{x,z}$ is onto and the dimension of Y is one more than that of W , this kernel is the line spanned by $y(x) = (1, g(q))$, where $g(q) := (\eta, \Theta, \Xi)$ is given by the equations above. If $p = \tau y(x) \in L_z(x)$ lies outside D , then

$$\text{dist}(p, D) \geq |\tau - 1|, \quad \text{dist}(p, K_z(x)) = |\tau - 1| \sqrt{1 + \|g(q)\|_2^2}.$$

Thus the non-tangency inequality holds for all these points with

$$S \geq \frac{1}{\sqrt{1 + \frac{4329}{16384}}} = \frac{128}{\sqrt{20713}} > 0.88.$$

Evaluating the reward at the unique point in $K_z(x)$ gives

$$v_z(x) = \frac{1}{2m} p_G(q).$$

Since q ranges over the radius- $\frac{3}{4}$ ball, homogeneity and Lemma 10 imply, for every $z \in Z$,

$$V_z = \frac{27}{128m} \max_{\|h\|_2=1} p_G(h) = \frac{27}{128m} \sqrt{\frac{2}{27}} \sqrt{1 - \frac{1}{\omega(G)}}.$$

For $j = 2, \dots, n$, let

$$U_j := \frac{27}{128m} \sqrt{\frac{2}{27}} \sqrt{1 - \frac{1}{j}}.$$

If $\omega(G) \geq k$, then $V_z \geq U_k$; if $\omega(G) \leq k-1$, then $V_z \leq U_{k-1}$. The gap between these bounds satisfies

$$U_k - U_{k-1} \geq \frac{27}{128m} \sqrt{\frac{2}{27}} \frac{1}{2k(k-1)} = \Omega\left(\frac{1}{mk^2}\right).$$

Since $\sqrt{\frac{2}{27}} > \frac{1}{4}$ and $m \leq n^2$, the gap $\Delta_k := U_k - U_{k-1}$ satisfies

$$\Delta_k > \frac{1}{40mk(k-1)} \geq \frac{1}{40mn^2}.$$

Set $\varepsilon := \frac{1}{500mn^2}$. Using polynomially many bits, we can compute a rational τ_k such that

$$\left| \tau_k - \frac{1}{2}(U_k + U_{k-1}) \right| \leq \varepsilon.$$

An estimate of V_z with additive error at most ε lies above τ_k when $\omega(G) \geq k$ and below it when $\omega(G) \leq k-1$, because $2\varepsilon < \frac{\Delta_k}{2}$. It therefore decides whether G contains a k -clique. The constructed instance has dimension $O(d^3)$, and all its defining coefficients are rational with polynomial encoding length. This proves the planning claim.

Given any rational arm, we can compute the unique point in $K_z(x)$ exactly in polynomial time using the equations above; $x_0 \geq \frac{2}{3}$ ensures that we can divide by x_0 . We can therefore simulate nature by returning this point. Applying Lemma 5 with the chosen ε gives an estimate of V_z with additive error at most ε , with probability at least $\frac{2}{3}$. Comparing it with τ_k decides CLIQUE, so $\text{NP} \subseteq \text{BPP}$. For a deterministic learner, the estimate is deterministic and gives $\text{P} = \text{NP}$. $\blacksquare$

E.E. Proof of Theorem 7

Suppose we are given rational descriptions of X , D , Z , the affine reward r , and the maps defining $K_z(x)$. For each hypothesis z , let

$$v_z(x) := \min_{y \in K_z(x)} r(x, y).$$

On its first round, IUCB's confidence set is all of Z , so its optimistic value is

$$V_{\text{IUCB}} := \max_{z \in Z, x \in X} v_z(x).$$

We will show that computing this value exactly is NP-hard. The same is true of finding an arm

$$x^* \in \operatorname{argmax}_{x \in X} \max_{z \in Z} v_z(x).$$

Proof Given a rational order-three tensor $\mathcal{T} \in \mathbb{Q}^{m \times d \times d_X}$, we can view it as a bilinear map $\mathcal{T} : \mathbb{R}^d \times \mathbb{R}^{d_X} \rightarrow \mathbb{R}^m$. Its spectral norm is

$$\|\mathcal{T}\|_\sigma := \max_{\|u\|_2 \leq 1, \|x\|_2 \leq 1} \|\mathcal{T}(u, x)\|_2.$$

Computing this value exactly is NP-hard [5]. We construct an IUCB instance from $\mathcal{T}$ so that an optimal first arm lets us recover this norm. Let

$$H := 1 + \sum_{i,j,k} |T_{ijk}|, \quad \alpha := \frac{1}{2H}, \quad X := \{x : \|x\|_2 \leq 1\}.$$

Write $z = (z_0, u)$ and $y = (w, s)$, and take

$$Z := \left\{ (z_0, u) : \left(z_0 - \frac{5}{3} \right)^2 + \|u\|_2^2 \leq 1 \right\}, \quad D := \left\{ (w, s) : \|w\|_2^2 + s^2 \leq 1 \right\}.$$

For $z = (z_0, u)$, define the constraint coefficients in Equation 1 and the reward by

$$B_z x := -\alpha \mathcal{T}(u, x), \quad C_z(w, s) := z_0 w, \quad d_z := 0, \quad r(x, (w, s)) := \frac{1}{2}(1 + s).$$

The feasible outcomes at arm x are then

$$K_z(x) = \left\{ (w, s) \in D : w = \frac{\alpha}{z_0} \mathcal{T}(u, x) \right\}.$$

The definition of Z gives $z_0 \geq \frac{2}{3}$ and

$$\frac{\|u\|_2}{z_0} \leq \frac{3}{4},$$

because $1 - \left(z_0 - \frac{5}{3}\right)^2 - 9\frac{z_0^2}{16} = -\frac{(15z_0-16)^2}{144} \leq 0$. Equality holds at $z_0 = \frac{16}{15}$ and $\|u\|_2 = \frac{4}{5}$ in every direction. Also, $\|\mathcal{T}(u, x)\|_2 \leq H\|u\|_2\|x\|_2$, so the constraint fixes w to a vector of norm at most $3\alpha\frac{H}{4} = \frac{3}{8}$. Thus every $K_z(x)$ is nonempty, and the uniform non-tangency condition holds with $S \geq \frac{\sqrt{55}}{8}$. To minimize the reward, nature chooses the smallest feasible s , namely $s = -\sqrt{1 - \|w\|_2^2}$. If we write $h(t) := \frac{1}{2}(1 - \sqrt{1 - t^2})$, then

$$v_z(x) = h\left(\frac{\alpha}{z_0} \|\mathcal{T}(u, x)\|_2\right).$$

For fixed z , define the matrix M_u by $M_{ux} := \mathcal{T}(u, x)$. A top right singular vector of M_u maximizes $v_z(x)$, so planning takes polynomial time when z is known. But IUCB must optimize over both z and x . Since h is strictly increasing and $\frac{\|u\|_2}{z_0}$ can equal $\frac{3}{4}$ in every direction,

$$V_{\text{IUCB}} = h\left(3\frac{\alpha}{4}\|\mathcal{T}\|_\sigma\right).$$

We can therefore recover the tensor norm from the optimistic value:

$$\|\mathcal{T}\|_\sigma = \frac{4}{3\alpha}\sqrt{1 - (1 - 2V_{\text{IUCB}})^2}.$$

If we are given only a globally optimal arm x^* , define the matrix N_x by $N_{xu} := \mathcal{T}(u, x^*)$. Computing its largest singular value gives

$$\max_{z \in Z} v_z(x^*) = h\left(3\frac{\alpha}{4}\sigma_{\max}(N_x)\right) = V_{\text{IUCB}}.$$

The same formula then recovers $\|\mathcal{T}\|_\sigma$, so finding a globally optimal first arm is NP-hard as well. ■

E.F. Proof of Theorem 8

An efficient planner for every hypothesis does not always give an efficient learner, so the converse of Lemma 5 fails. We show this with finite arm and hypothesis sets that have short descriptions. We also allow the reward to be linear in the arm and in the outcome separately, so it can contain products of arm and outcome coordinates.

Proof Fix $2 \leq k \leq n$, and let

$$E_n := \{\{i, j\} : 1 \leq i < j \leq n\}, \quad m := \binom{n}{2}, \quad q := \binom{k}{2}.$$

For every k -element set $T \subseteq [n]$, define $x^T \in \mathbb{R}^m$ by

$$x_e^T := \begin{cases} \frac{1}{q} & \text{if } e \subseteq T \\ 0 & \text{otherwise} \end{cases}, \quad e \in E_n,$$

and take

$$X := \{x^T : T \subseteq [n], |T| = k\}.$$

An arm is represented succinctly by the vertex set T . For every k -element set $U \subseteq [n]$, define $z^U \in \mathbb{R}^{1+m}$ by

$$z_0^U := 1, \quad z_e^U := \begin{cases} 1 & \text{if } e \subseteq U \\ 0 & \text{otherwise} \end{cases},$$

and let $Z := \{z^U : U \subseteq [n], |U| = k\}$.

Write an outcome as $y = (y_0, g, s) \in \mathbb{R}^{1+2m}$ and set

$$D := \{(1, g, s) : g \in [0, 1]^m, s \in [-1, 1]^m\}.$$

With constraint space $W = \mathbb{R}^m$, let $F_0(x, z) := 0$ and define

$$(F_1(y, z))_e := z_e(g_e - y_0 - s_e) + z_0 s_e, \quad e \in E_n.$$

For $z \in Z$, set

$$K_z(x) := \{y \in D : F_0(x, z) + F_1(y, z) = 0\}, \quad h_z(x) := \{Q \in \Delta D : \mathbb{E}_{y \sim Q}[y] \in K_z(x)\},$$

and let $H := \{h_z : z \in Z\}$.

The maps F_0 and F_1 are bilinear in their displayed arguments. Since y_0 is an outcome coordinate, we can write the constraint as $C_z y = 0$, with $B_z = 0$, $d_z = 0$, and $z \mapsto C_z$ linear. For $z = z^U$,

$$(F_1(y, z^U))_e = \begin{cases} g_e - y_0 & \text{if } e \subseteq U \\ s_e & \text{otherwise.} \end{cases}$$

Thus, inside D , the constraints force $g_e = 1$ for every $e \subseteq U$ and $s_e = 0$ for every $e \not\subseteq U$. The remaining g_e coordinates are free to vary in $[0, 1]$. The feasible set is nonempty: take $g_e = 1$ on the edges of U , $g_e = 0$ elsewhere, and $s = 0$. Moreover, $y \mapsto F_1(y, z^U)$ is onto W : given $w \in W$, set $y_0 = 0$, use $g_e = w_e$ and $s_e = 0$ on the edges of U , and use $g_e = 0$ and $s_e = w_e$ elsewhere.

Define the reward

$$r(x, (y_0, g, s)) := \sum_{e \in E_n} x_e g_e.$$

The learner is given a succinct description of the entire instance $\mathcal{M} = (X, D, H, r)$, including Z , F_0 , and F_1 ; only the true U is hidden. The reward lies in $[0, 1]$ on $X \times D$. For an arm x^T , nature minimizes it by setting every graph coordinate not forced by U to zero, so

$$v_{z^U}(x^T) = \frac{\binom{|U \cap T|}{2}}{q}.$$

Thus $V_{z^U} = 1$, uniquely attained by x^U . Given z^U , an exact planner returns x^U by setting $x_e^U = \frac{z_e^U}{q}$ for every $e \in E_n$, which takes $O(m) = O(n^2)$ time.

We now show that a learner with the stated regret bound would solve CLIQUE. Given a graph $G = ([n], E(G))$ and target clique size k [6], let

$$g_e^G := \begin{cases} 1 & \text{if } e \in E(G) \\ 0 & \text{otherwise} \end{cases}, \quad y^G := (1, g^G, 0) \in D.$$

If U is a k -clique of G , nature can return y^G on every round under hypothesis z^U : every edge inside U has $g_e^G = 1$, and all the s_e coordinates are zero. For every candidate T ,

$$r(x^T, y^G) = \frac{|E(G[T])|}{q}.$$

This reward is 1 exactly when T is a k -clique, and is at most $1 - \frac{1}{q}$ otherwise.

Assume the policy in the theorem exists, and give it the outcome y^G on every round. If G contains a k -clique U , these outcomes are compatible with z^U , whose optimal value is 1. If the policy plays $x^{T_1}, \dots, x^{T_N}$, the total reward it loses relative to this value is

$$\mathcal{L}_N := N - \sum_{t=1}^N r(x^{T_t}, y^G).$$

This quantity is nonnegative. Let $\mathcal{E}$ be the event that none of $T_1, \dots, T_N$ is a clique. On this event, the learner loses at least $\frac{1}{q}$ on every round, so $\mathcal{L}_N \geq \frac{N}{q}$. Markov's inequality and the assumed regret bound give

$$\Pr(\mathcal{E}) \leq q \frac{E[\mathcal{L}_N]}{N} \leq qP(|\mathcal{M}|) N^{-\beta}.$$

Choose

$$N := \left\lceil (4qP(|\mathcal{M}|))^{\frac{1}{\beta}} \right\rceil.$$

The instance description has length polynomial in n , so N and the time needed to simulate the learner are polynomial in the CLIQUE input length. Check each k -set proposed by the learner and accept as soon as one is a clique in G . If G has a k -clique, the preceding bound gives acceptance probability at least $\frac{3}{4}$. If G has none, none of the proposed sets can pass this check. In that case, the outcomes may be incompatible with every hypothesis, but the learner must still terminate in polynomial time on every syntactically valid history. Thus the simulation also takes polynomial time when G has no k -clique.

This is an RP algorithm for CLIQUE. Since CLIQUE is NP-complete, it implies $\text{NP} = \text{RP}$. If the policy is deterministic, the same verified search gives $\text{P} = \text{NP}$. ■

This example shows that efficient planning alone does not suffice for efficient learning. It uses exponentially large finite sets X and Z represented by k -subsets, a polytope D , and the bilinear reward $r(x, y) = \sum_e x_e g_e$. It therefore falls outside our setting of Euclidean balls and jointly affine rewards.